

Programme	B. Sc. Mathematics Honours			
Course Code	MAT6EJ304 (2)			
Course Title	MACHINE LEARNING - II			
Type of Course	Elective (Specialisation- Data Science)			
Semester	VI			
Academic Level	300 - 399			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Machine Learning - I			
Course Summary	This course studies advanced models of machine learning. Mastery of techniques like regression, classification, and dimensionality reduction will enable students to handle complex data sets, perform advanced analytics, and develop robust predictive models. Understanding kernel methods, SVMs, graphical models, and PCA will provide the necessary tools for tackling a wide range of data-driven challenges in real-world applications.			

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To analyse and design support vector machines and kernel methods for learning problem.	An	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO2	To analyse graphical models for learning and explore belief propagation in graph models.	An	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	To analyse and apply PCA and dimensionality reduction techniques	Ap	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book		Pattern Recognition and Machine Learning - Christopher M. Bishop - Springer - 2006		
Module	Unit	Content	Hrs (48+12)	Ext. Marks (70)
I	Kernel Methods		12	
	1	Review of linear regression and classification		
	2	Dual representations and construction of kernels		
	3	Radial basis function networks - Nadaraya-Watson model		
	4	Gaussian processes for regression and classification		
	5	Laplace approximation		
	6	Connection to neural networks		
	Chapter 6 of text book			
II	Support Vector Machines		12	
	7	Maximum Margin Classifiers		
	8	Relation to logistic regression		
	9	Regression using SVM.		
	10	Relevance Vector Machines		
	11	Regression and classification using RVM		
	Chapter 7 of text book			
III	Graphical Models		12	
	12	Bayesian Networks		
	13	Markov Random Fields		
	14	Factorization properties		
	15	Inference in Graphical Models		
	16	Factor graphs and sum-products algorithm		
	17	Belief propagation		
	Chapter 8 of text book			
IV	Principal Component Analysis		12	
	18	Maximum variance and minimum error PCA		

	19	Dimensionality reduction		
	20	Maximum likelihood PCA and EM algorithm		
	21	Bayesian PCA and factor analysis		
	22	Kernel PCA		
	Chapter 12 of text book			
V	Open Ended		12	
		1. Boosting 2. Convex learning problems 3. Regularization in convex learning 4. Learning of convex Lipschitz and smooth bounded functions 5. Stochastic gradient descent		
References:				
1)Understanding Machine Learning from Theory to Algorithms - Shai Shalev Shwartz, Shai Ben David - Cambridge University Press - ISBN 978-1-107-05713-5 - 2014				
2) Foundations of Machine Learning - Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar - The MIT Press - 2012				

Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	3	2	2	3	2	3	2	3	3	3	1	3
CO 2	3	3	2	2	3	2	3	2	3	3	3	1	3
CO 3	3	3	2	2	3	2	3	2	3	3	3	1	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓